

Knowledge Representation

Logical Agents

Chapter 7

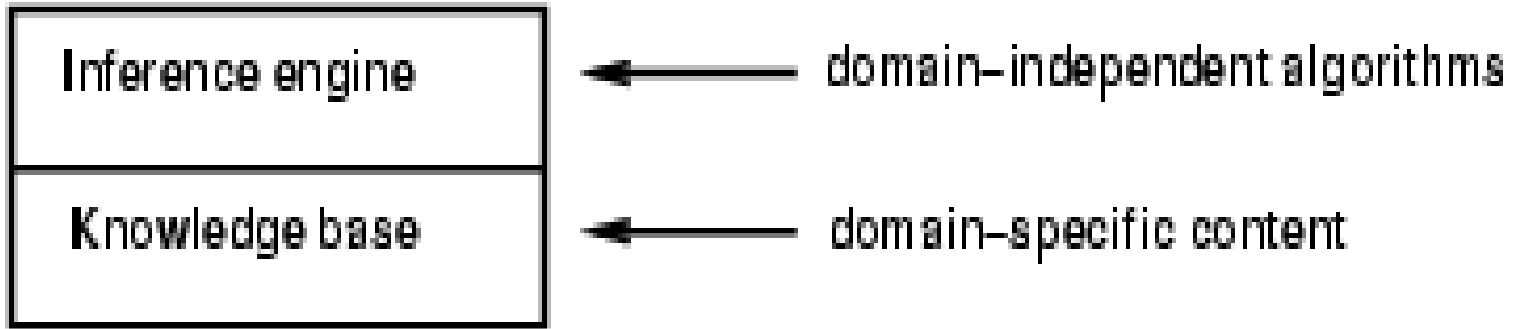
Outline

- Knowledge-based agents
- Wumpus world
- Logic in general - models and entailment
- Propositional (Boolean) logic
- Equivalence, validity, satisfiability
- Inference rules and theorem proving
 - forward chaining
 - backward chaining
 - resolution

Why knowledge base?

- Knowledge and reasoning
- It enable successful behaviors of an intelligent agents
- In partially observable environment - a KB agent can combine general knowledge with current percept to infer hidden aspect of the current state prior to selecting actions
- Ex: Physician, Natural language processing
- E.g. John threw the brick through window and broke it.
- Flexibility :
 1. Agents are able to accept new task
 2. Agents can achieve competence quickly by being told or learning new knowledge,
 3. Agents can adapt to changes in the environment by updating the relevant knowledge

Knowledge bases



- Knowledge base = set of **sentences** in a **formal** language
- **Declarative** approach to building an agent (or other system):
 - Tell it what it needs to know
- Then it can Ask itself what to do - answers should follow from the KB
- Agents can be viewed at the **knowledge level**
i.e., what they know, regardless of how implemented
- Or at the **implementation level**
 - i.e., data structures in KB and algorithms that manipulate them

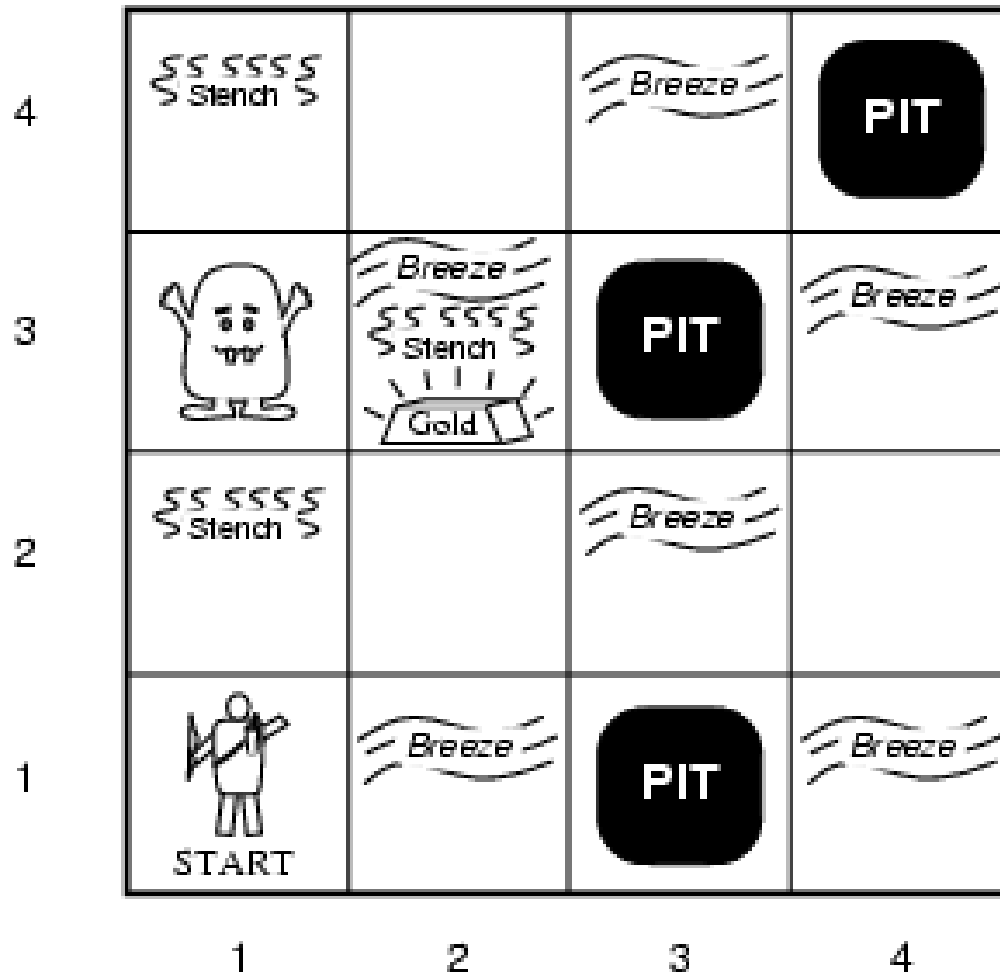
A simple knowledge-based agent

```
function KB-AGENT(percept) returns an action
  static: KB, a knowledge base
          t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))
  action  $\leftarrow$  ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))
  t  $\leftarrow$  t + 1
  return action
```

- The agent must be able to:
 - Represent states, actions, etc.
 - Incorporate new percepts
 - Update internal representations of the world
 - Deduce hidden properties of the world
 - Deduce appropriate actions

Wumpus World PEAS description

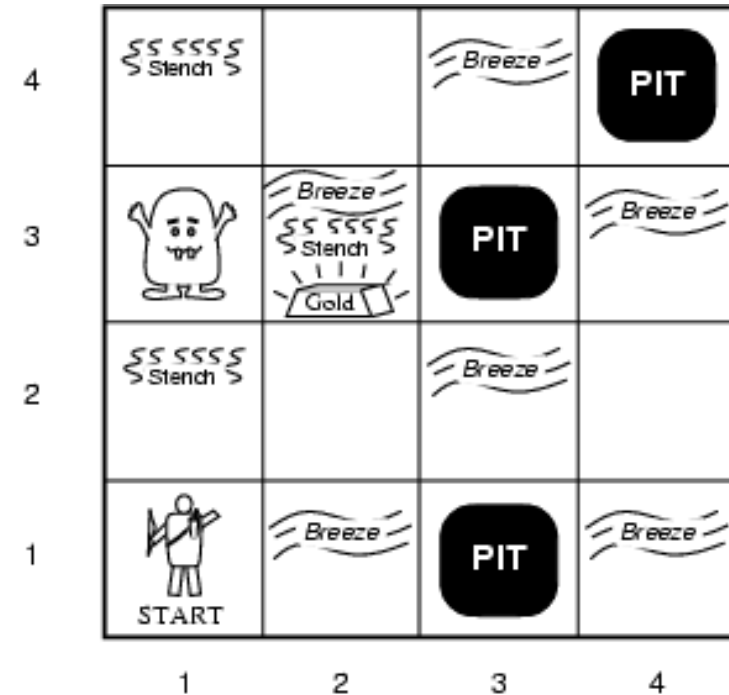


- Performance measure
 - gold +1000, death -1000
 - -1 per step, -10 for using the arrow

Wumpus World PEAS description

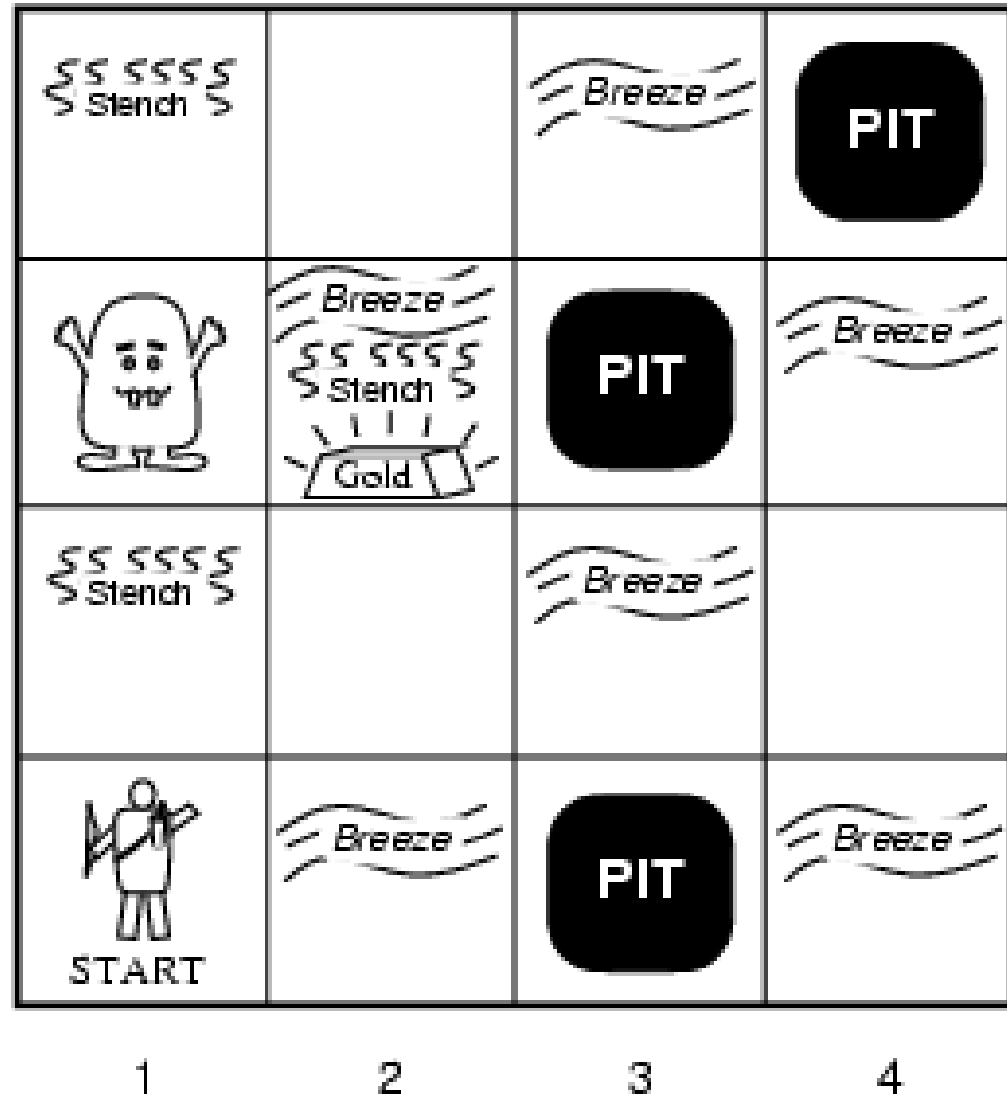
- Environment

- 4*4 grid
- Agent always starts in [1,1], facing to the right
- Location of wumpus and gold are chosen randomly with uniform distribution from squares other than start square
- Each square other than start can be a pit with probability 0.2
- Squares adjacent (not diagonally) to wumpus are smelly
- Squares adjacent (not diagonally) to pit are breezy
- Glitter iff gold is in the same square
- Shooting kills wumpus if you are facing it
- Shooting uses up the only arrow
- Grabbing picks up gold if in same square



Wumpus World PEAS description

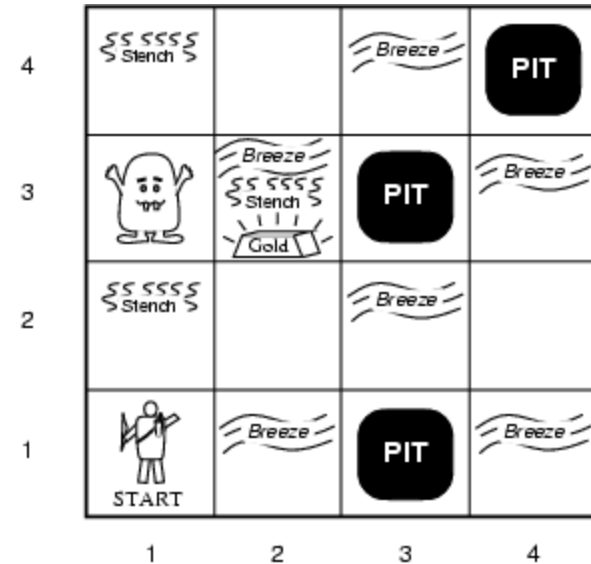
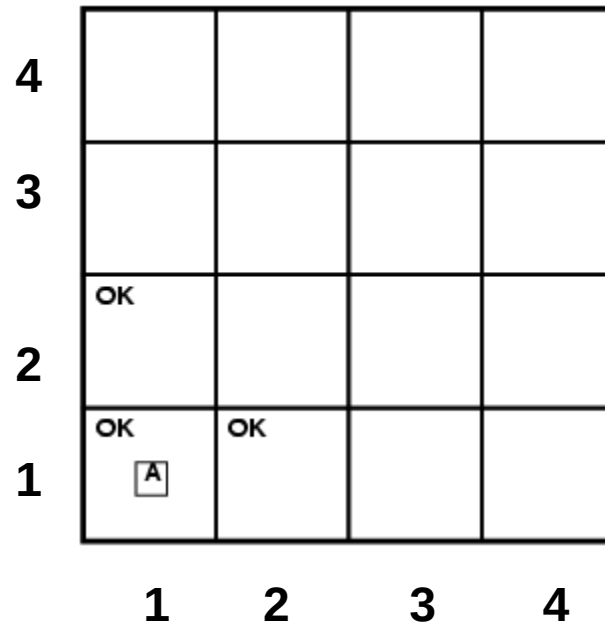
- **Sensors:**
 - Stench,
 - Breeze,
 - Glitter,
 - Bump (when agent walks into a wall),
 - Scream (when wumpus is killed)
- **Actuators:** Left turn, Right turn, Forward, Grab, Release, Shoot
- The percept is given in the form of symbols [St, Br, G, Bu, Sc]



Wumpus world characterization

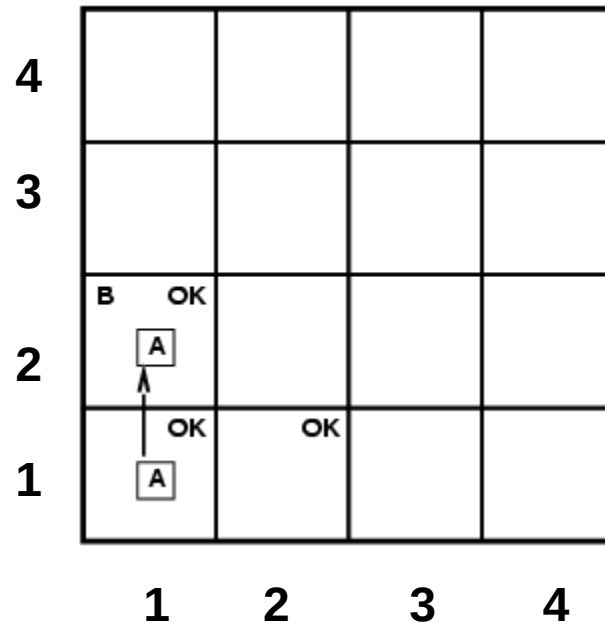
- Fully_Observable No – only **local** perception
- Deterministic Yes – outcomes exactly specified
- Episodic No – sequential at the level of actions
- Static Yes – Wumpus and Pits do not move
- Discrete Yes
- Single-agent? Yes – Wumpus is essentially a natural feature

Exploring a wumpus world

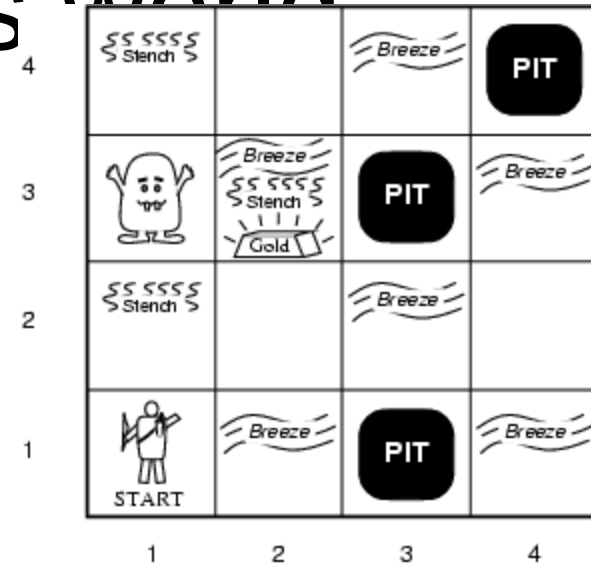


First percept [None,N,N,N,N]

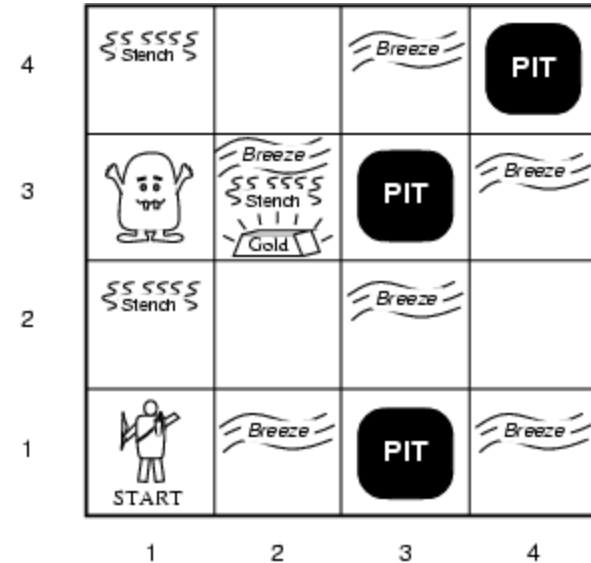
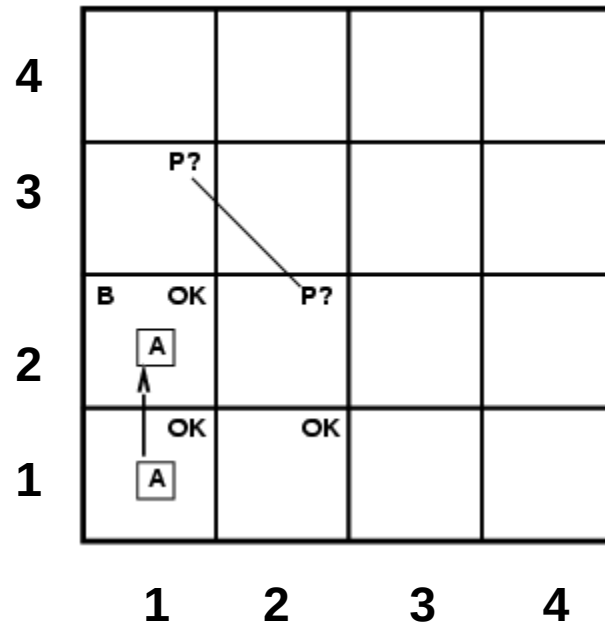
Exploring a wumpus world



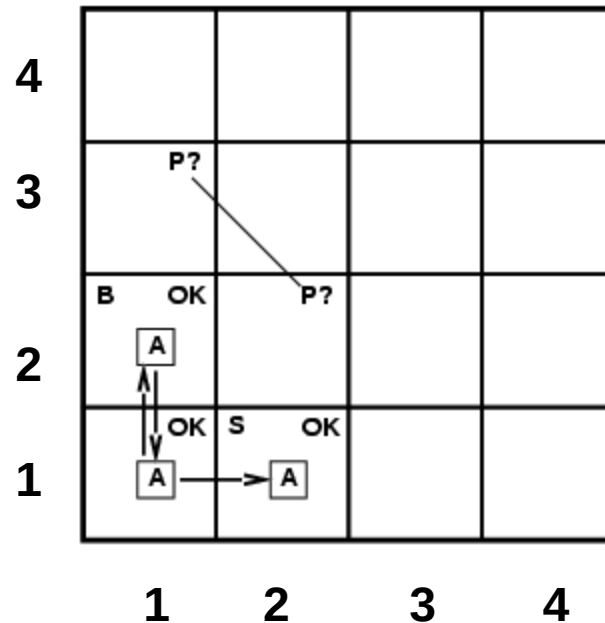
Next percept after one move
[None, Breeze, N, N, N]



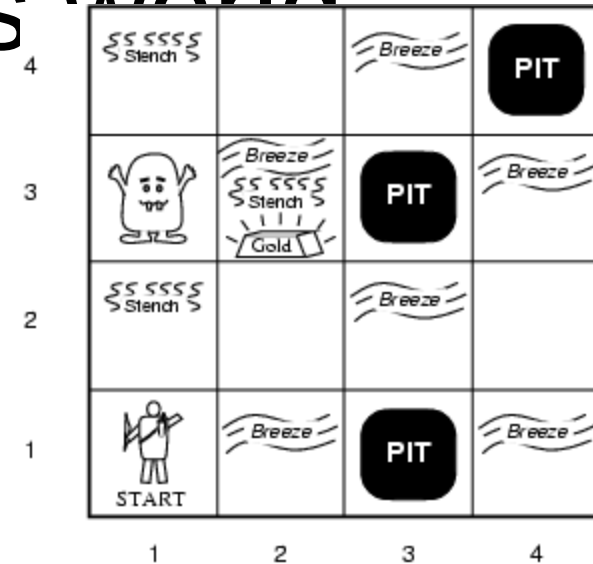
Exploring a wumpus world



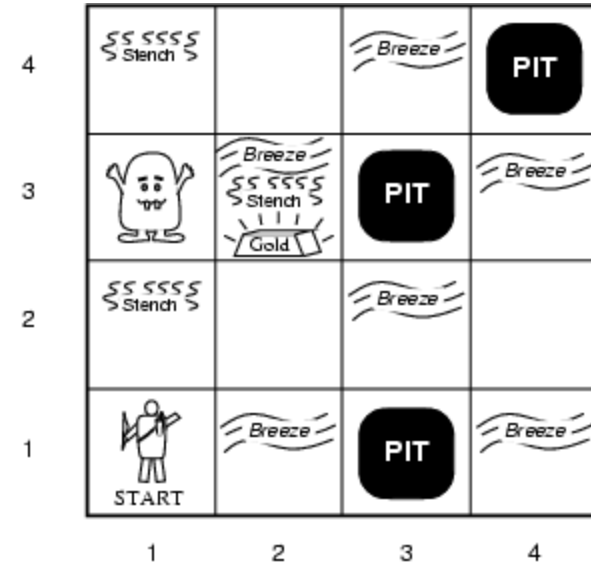
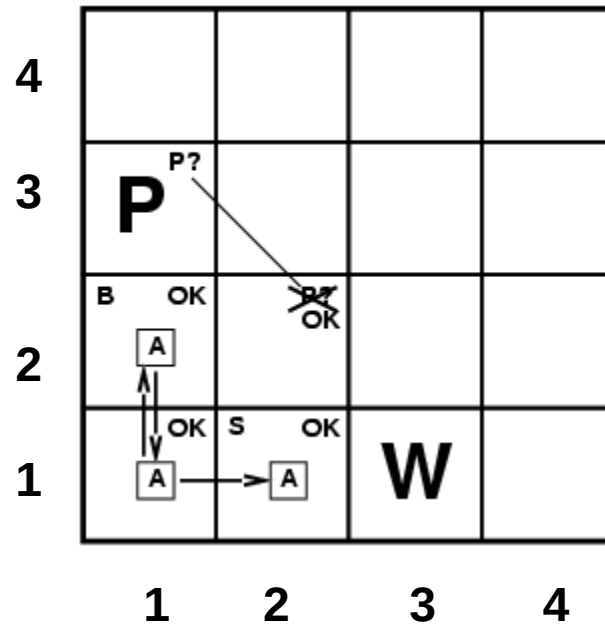
Exploring a wumpus world



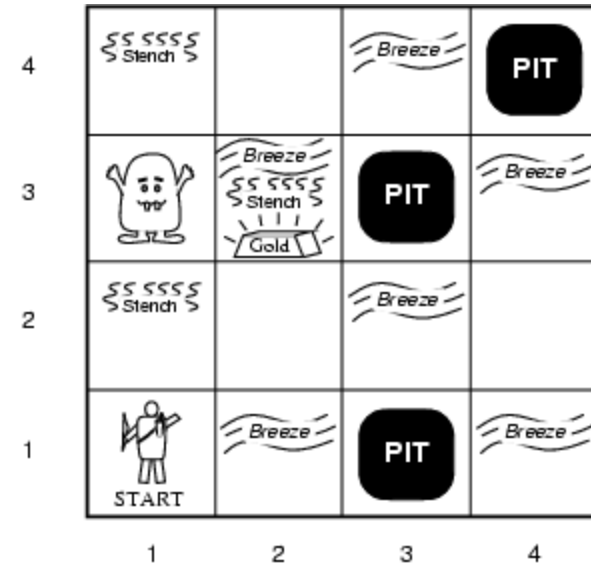
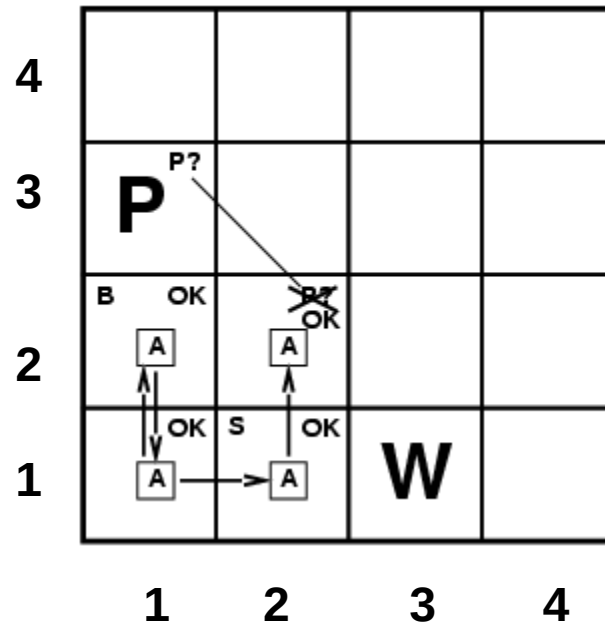
The new percept in [2,1] is
[Stench, None, N, N, N]



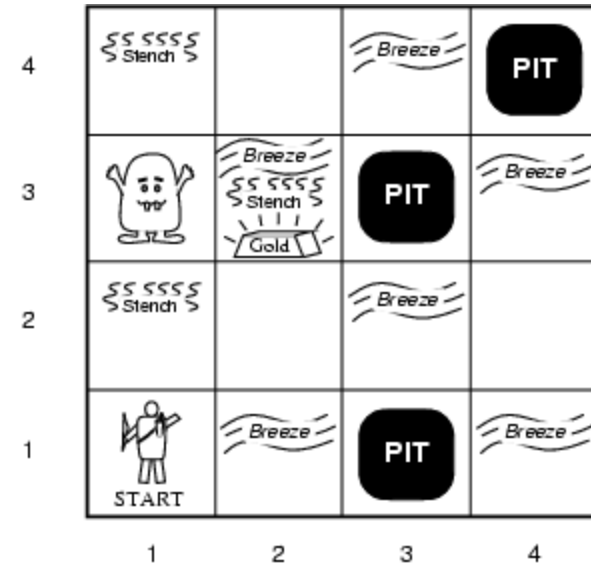
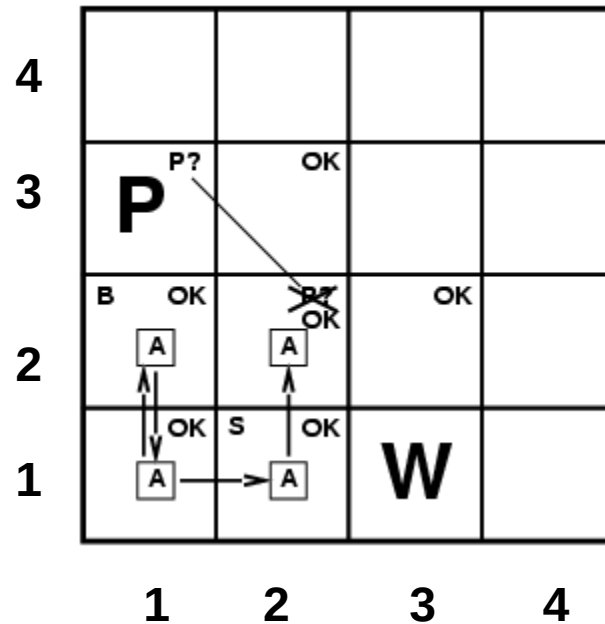
Exploring a wumpus world



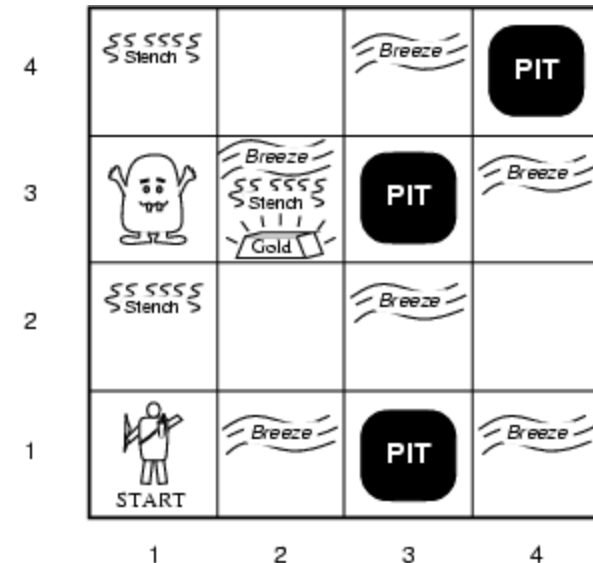
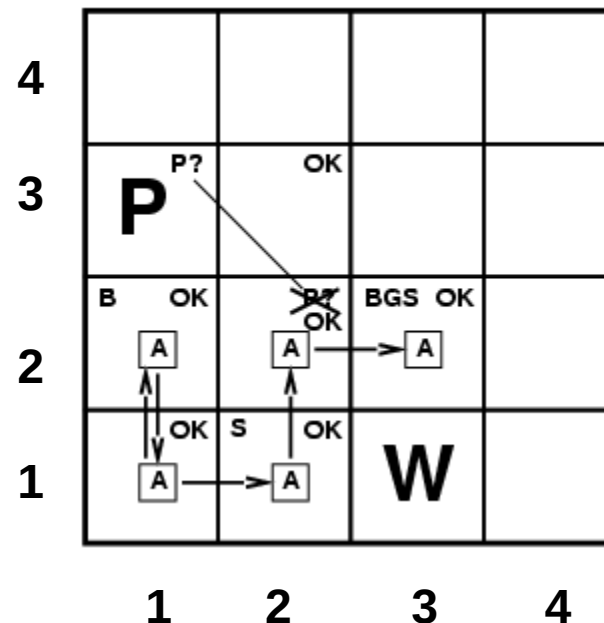
Exploring a wumpus world



Exploring a wumpus world



Exploring a wumpus world



In each case where the agent draws a conclusion from the available information, that conclusion is guaranteed to be correct if the available information is correct

Logic in general

- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax** defines the sentences in the language
- **Semantics** define the "meaning" of sentences;
 - i.e., define **truth** of a sentence in a world
- E.g., the language of arithmetic
 - $x+2 \geq y$ is a sentence; $x^2+y > \{ \}$ is not a sentence
 - $x+2 \geq y$ is true iff the number $x+2$ is no less than the number y
 - $x+2 \geq y$ is true in a world where $x = 7, y = 1$
 - $x+2 \geq y$ is false in a world where $x = 0, y = 6$

Representing Knowledge: Propositional Logic

- Symbols
- *True, False*
- Implication: $\Rightarrow$
- Equivalence: $\Leftrightarrow$
- And (conjunction): $\wedge$
- Or (disjunction): $\vee$ Negation: $\neg$
- Sentences = combination of the symbols, truth values, and operators
- Literals = Symbols and negated symbols (A and $\neg A$ are both literals)
 - Raining $\Rightarrow$ Wet
 - $\neg(\text{Busy} \wedge \text{Sleeping})$
 - $(A \wedge B) \vee \neg C$

Knowledge Base (KB)

- **Knowledge Base (KB):** a collection of sentences.
- **Model:** an assignment of (*True/False*) values to each of the symbols. If the knowledge base is built from n symbols, there are 2^n possible models.
- **Evaluation:** A sentence s is evaluated on a model m by setting each symbol to its corresponding value in m . The result of the evaluation is a value in $\{\text{True}, \text{False}\}$
- **KB Evaluation:** The result of the KB evaluation is the conjunction of the results of the evaluations of all the sentences in KB

Example KB

KB:

$A \vee B$

$\neg C \vee A$

<i>A</i>	<i>B</i>	<i>C</i>	<i>KB</i>
<i>F</i>	<i>F</i>	<i>F</i>	<i>F</i>
<i>F</i>	<i>F</i>	<i>T</i>	<i>F</i>
<i>F</i>	<i>T</i>	<i>F</i>	<i>T</i>
<i>F</i>	<i>T</i>	<i>T</i>	<i>F</i>
<i>T</i>	<i>F</i>	<i>F</i>	<i>T</i>
<i>T</i>	<i>F</i>	<i>T</i>	<i>T</i>
<i>T</i>	<i>T</i>	<i>F</i>	<i>T</i>
<i>T</i>	<i>T</i>	<i>T</i>	<i>T</i>

Logical Entailment

- “KB logically entails *S*” if all the models that evaluate KB to *True* also evaluate *S* to *True*.
- Denoted by: $\text{KB} \models S$
- Note: We do not care about those models that evaluate KB to *False*. The result of evaluating *S* for these models is irrelevant.

Q.1 Does KB entails $A \wedge C$

Q.2 Does KB entails $A \vee B \vee C$

Example KB

KB:

$$A \vee B$$

$$\neg C \vee A$$

S:

$$A \wedge C$$

A	B	C	KB	S
F	F	F	F	F
F	F	T	F	F
F	T	F	T	F
F	T	T	F	F
T	F	F	T	F
T	F	T	T	T
T	T	F	T	F
T	T	T	T	T

~~KB $\models$ S~~
because
KB is true
but S is
false

Example KB

KB:

$$A \vee B$$

$$\neg C \vee A$$

S:

$$A \vee B \vee C$$

A	B	C	KB	S
F	F	F	F	F
F	F	T	F	T
F	T	F	T	T
F	T	T	F	T
T	F	F	T	T
T	F	T	T	T
T	T	F	T	T
T	T	T	T	T

KB $\models$ S
because S is true for
all the assignments for
which KB is true

Entailment

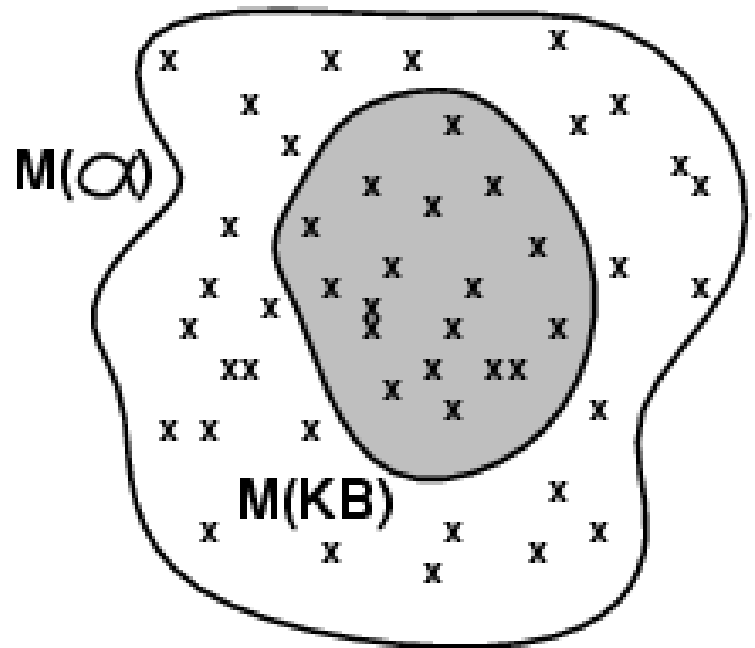
- **Entailment** means that one thing **follows from** another:

$$KB \models \alpha$$

- Knowledge base KB entails sentence α if and only if α is true in all worlds where KB is true
 - E.g., the KB containing “the Giants won” and “the Reds won” entails “Either the Giants won or the Reds won”
 - E.g., $x+y = 4$ entails $4 = x+y$
 - Entailment is a relationship between sentences (i.e., **syntax**) that is based on **semantics**

Models

- Logicians typically think in terms of **models**, which are formally structured worlds with respect to which truth can be evaluated
-
- We say m is a model of a sentence α if α is true in m
- $M(\alpha)$ is the set of all models of α
-
- Then $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$
- - E.g. $KB = \text{Giants won and Reds won}$ $\alpha = \text{Giants won}$
 -

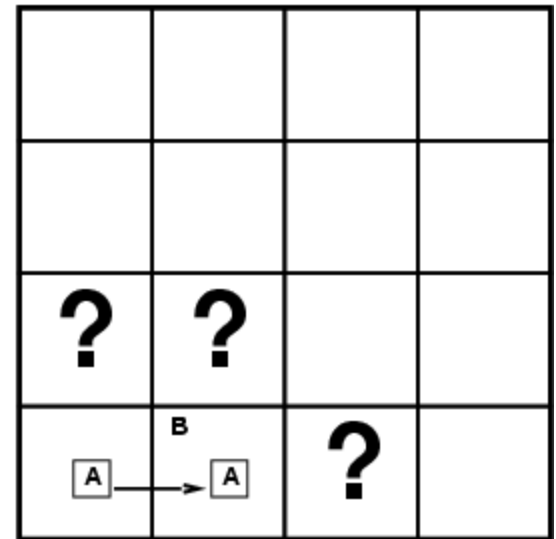


Entailment in the wumpus world

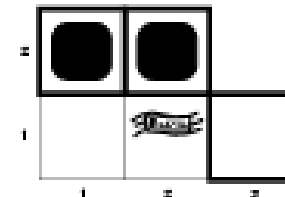
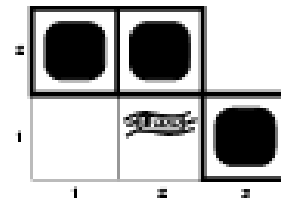
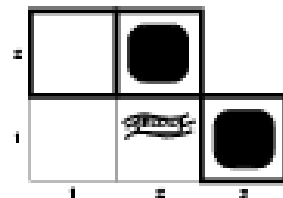
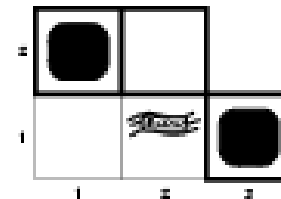
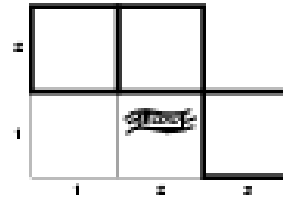
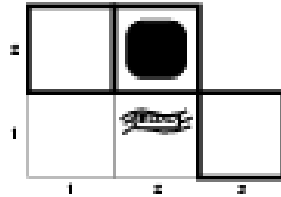
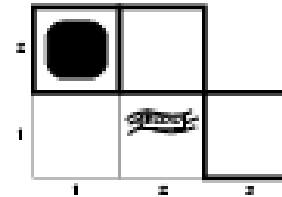
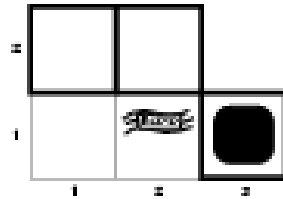
Situation after detecting
nothing in [1,1], moving
right, breeze in [2,1]

Consider possible models for
KB assuming only pits

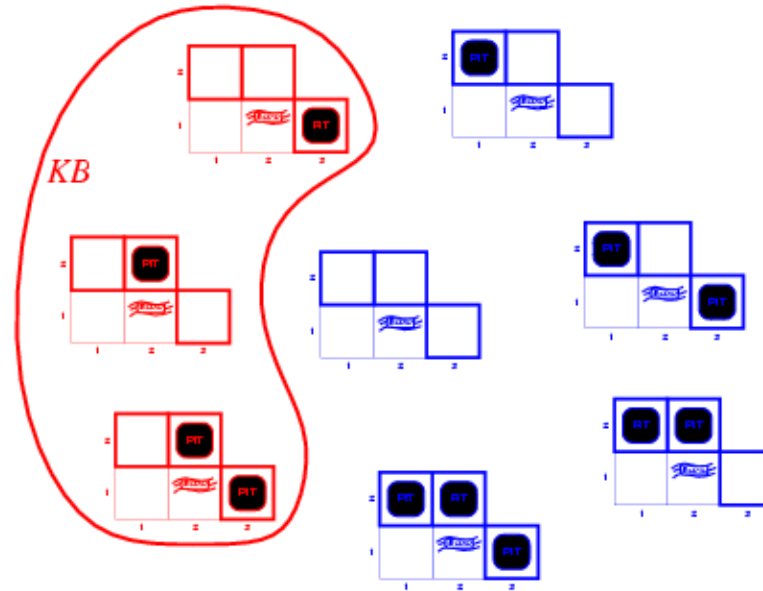
3 Boolean choices $\Rightarrow$ 8
possible models



Wumpus models

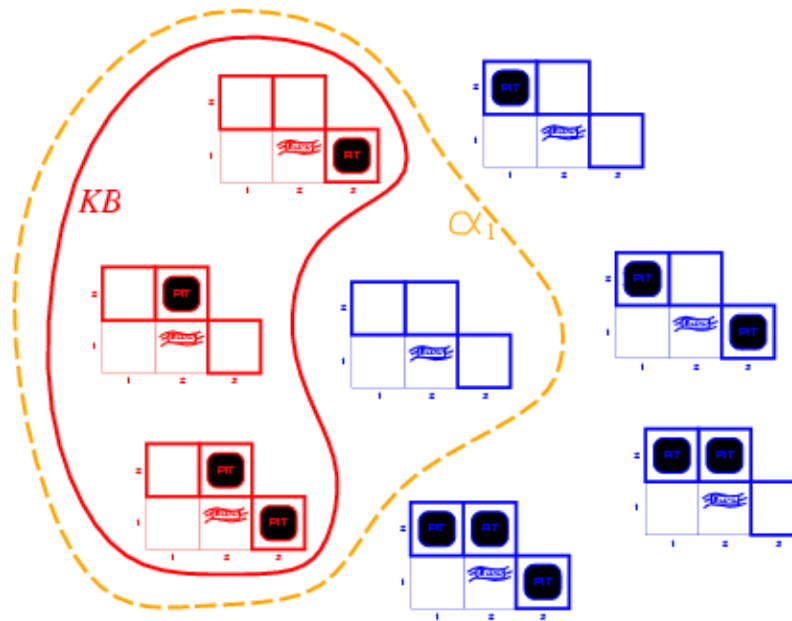


Wumpus models



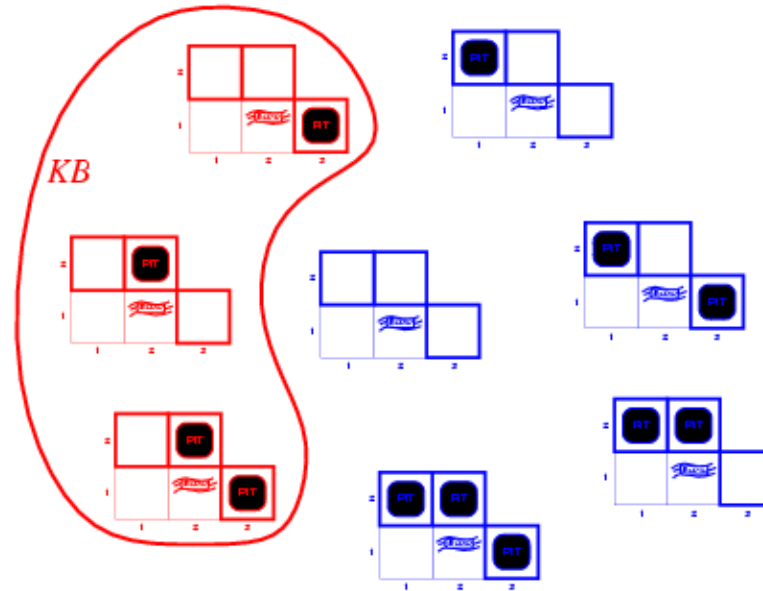
- KB = wumpus-world rules + observations
-

Wumpus models



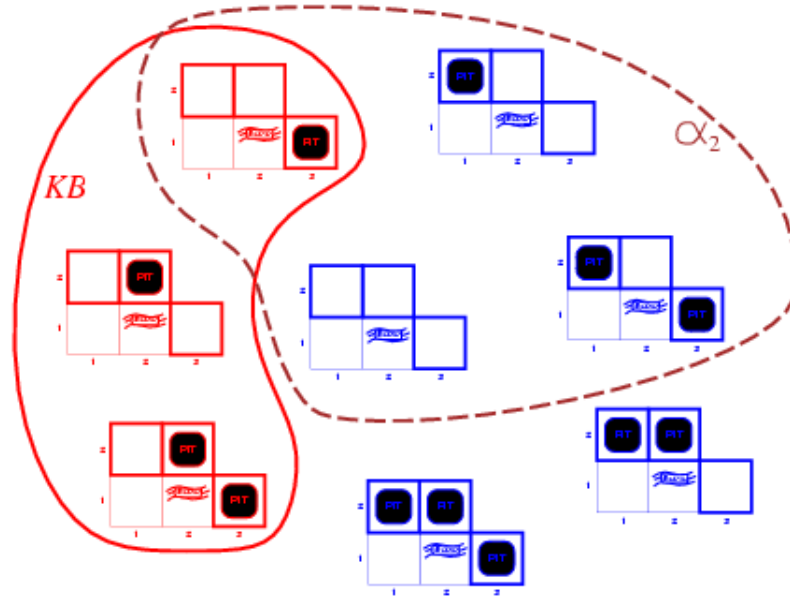
- KB = wumpus-world rules + observations
- α_1 = "[1,2] is safe", $KB \models \alpha_1$, proved by model checking

Wumpus models



- KB = wumpus-world rules + observations

Wumpus models



- KB = wumpus-world rules + observations
- α_2 = "[2,2] is safe", $KB \not\models \alpha_2$
-

Inference

- $KB \vdash_i \alpha$ = sentence α can be derived from KB by procedure i
- **Soundness**: i is sound if whenever $KB \vdash_i \alpha$, it is also true that $KB \models \alpha$ (i.e. derives only entailed sentence)
- i.e. it produces only true statements
- **Completeness**: i is complete if whenever $KB \models \alpha$, it is also true that $KB \vdash_i \alpha$
- i.e. it produces all true statements
- Preview: we will define a logic (first-order logic) which is expressive enough to say almost anything of interest, and for which there exists a sound and complete inference procedure.
- That is, the procedure will answer any question whose answer follows from what is known by the KB .

Propositional logic: Syntax

- Propositional logic is the simplest logic – illustrates basic ideas
- The proposition symbols P_1 , P_2 etc are sentences
 - If S is a sentence, $\neg S$ is a sentence (negation)
 -
 - If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (conjunction)
 -
 - If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (disjunction)
 -
 - If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (implication)
 -
 - If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (biconditional)
 -

Propositional logic: Semantics

Each model specifies true/false for each proposition symbol

E.g. $P_{1,2}$ $P_{2,2}$ $P_{3,1}$
false true false

With these symbols, 8 possible models, can be enumerated automatically.

Rules for evaluating truth with respect to a model m :

$\neg S$	is true iff	S is false
$S_1 \wedge S_2$	is true iff	S_1 is true and S_2 is true
$S_1 \vee S_2$	is true iff	S_1 is true or S_2 is true
$S_1 \Rightarrow S_2$	is true iff	S_1 is false or S_2 is true
i.e.,	is false iff	S_1 is true and S_2 is false
$S_1 \Leftrightarrow S_2$	is true iff	$S_1 \Rightarrow S_2$ is true and $S_2 \Rightarrow S_1$ is true

Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$$

Truth tables for connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

- Using this table, the truth value of any sentence s can be computed w. r. t. any model m by a simple process of recursive evaluation

A simple knowledge base - Wumpus world sentences

Need to choose the vocabulary of Prop. Symbols

- Let $P_{i,j}$ be true if there is a pit in $[i, j]$.
- Let $B_{i,j}$ be true if there is a breeze in $[i, j]$.

Knowledge base include the following sentences

- There is no pit in $[1,1]$

$$R1 : \neg P_{1,1}$$

- "Pits cause breezes in adjacent squares"

$$R2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$R3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

The percept for first two square visited by an agent

$$R4 : \neg B_{1,1}$$

$$R5 : B_{2,1}$$

Truth tables for inference

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	KB	α_1
<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>true</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>

- The relevant symbols are
- 128 possible models

Validity and satisfiability

A sentence is **valid** if it is true in **all** models,
e.g., *True*, $A \vee \neg A$, $A \Rightarrow A$, $(A \wedge (A \Rightarrow B)) \Rightarrow B$

Validity is connected to inference via the **Deduction Theorem**:
 $KB \models \alpha$ if and only if $(KB \Rightarrow \alpha)$ is valid

A sentence is **satisfiable** if it is true in **some** model
e.g., $A \vee B$, C

A sentence is **unsatisfiable** if it is true in **no** models
e.g., $A \wedge \neg A$

Satisfiability is connected to inference via the following:
 $KB \models \alpha$ if and only if $(KB \wedge \neg \alpha)$ is unsatisfiable

Convert following sentences into propositional logic

1. Alice likes maths and she likes stories
2. If she likes maths, she likes algebra
3. If she likes algebra and likes physics she will go to college
4. She does not like stories or she likes physics
5. She does not like chemistry and history

Propositional logic

- P = Alice likes maths
- Q = Alice likes stories
- R = Alice likes algebra
- S = Alice likes physics
- T = Alice will go to college
- U = Alice likes chemistry
- V = Alice like history

hypothesis

- $P \wedge Q$
- $P \rightarrow R$
- $(R \wedge S) \rightarrow T$
- $\neg Q \vee S$
- $\neg U \wedge \neg V$

Rule of inference	Tautology	Name
$\begin{array}{l} p \rightarrow q \\ \underline{p} \\ \therefore q \end{array}$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ \underline{p \rightarrow q} \\ \therefore \neg p \end{array}$	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ \underline{q \rightarrow r} \\ \therefore p \rightarrow r \end{array}$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \underline{\neg p} \\ \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} \underline{p \wedge q} \\ \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \underline{\neg p \vee r} \\ \therefore q \vee r \end{array}$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (p \vee r)$	Resolution

Soundness of Resolution: Validity of the Resolution Inference Rule

$$\begin{array}{l} P \vee Q \\ \neg P \vee R \\ \hline \therefore Q \vee R \end{array}$$

resolving on P

Validity (Tautology): $(P \vee Q) (\neg P \vee R) \rightarrow (Q \vee R) ;$

P	Q	R	$(P \vee Q)$	$(\neg P \vee R)$	$(P \vee Q) \wedge (\neg P \vee R)$	$(Q \vee R)$	$(P \vee Q) (\neg P \vee R) \rightarrow (Q \vee R)$
0	0	0	0	1	0	0	1
0	0	1	0	1	0	1	1
0	1	0	1	1	1	1	1
1	0	0	1	0	0	0	1
1	1	0	1	0	0	1	1
1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1
0	0	0	0	1	0	0	1

Another view on what we are doing

You might think of this as some sort of game.

You are given some statement, and you want to see if it is a valid argument and true

You translate the statement into argument form using propositional variables, and make sure you have the premises right, and clear what is the conclusion

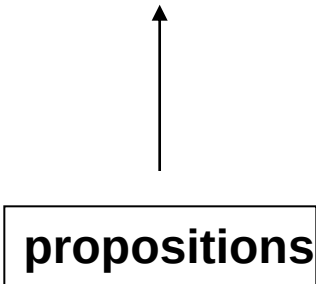
You then want to get from premises/hypotheses (A) to the conclusion (B) using the rules of inference.

So, get from A to B using as “moves” the rules of inference

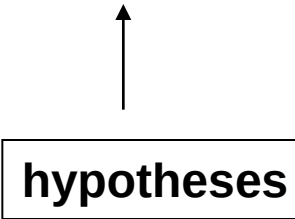
**It is not sunny this afternoon and it is colder than yesterday.
If we go swimming it is sunny.
If we do not go swimming then we will take a canoe trip.
If we take a canoe trip then we will be home by sunset.
We will be home by sunset**

- 1. It is not sunny this afternoon and it is colder than yesterday.
- 2. If we go swimming it is sunny.
- 3. If we do not go swimming then we will take a canoe trip.
- 4. If we take a canoe trip then we will be home by sunset.
- 5. We will be home by sunset

p It is sunny this afternoon
 q It is colder than yesterday
 r We go swimming
 s We will take a canoe trip
 t We will be home by sunset (the conclusion)



- 1. $\neg p \wedge q$
- 2. $r \rightarrow p$
- 3. $\neg r \rightarrow s$
- 4. $s \rightarrow t$
- 5. t



Using the rules of inference to build arguments

An example

- p It is sunny this afternoon
 q It is colder than yesterday
 r We go swimming
 s We will take a canoe trip
 t We will be home by sunset (the conclusion)

1. $\neg p \wedge q$
2. $r \rightarrow p$
3. $\neg r \rightarrow s$
4. $s \rightarrow t$
5. t

Step	Reason
1. $\neg p \wedge q$	Hypothesis
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Hypothesis
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Hypothesis
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Hypothesis
8. t	Modus ponens using (6) and (7)

Rule of inference	Tautology	Name
$\frac{p \rightarrow q \quad p}{\therefore q}$	$[p \wedge (p \rightarrow q)] \rightarrow q$	Modus ponens
$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$	Modus tollens
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$	Resolution

Using the resolution rule (an example)

1. Anna is skiing or it is not snowing.
2. It is snowing or Bart is playing hockey.
3. Consequently Anna is skiing or Bart is playing hockey.

We want to show that (3) follows from (1) and (2)

Using the resolution rule (an example)

1. Anna is skiing or it is not snowing.
2. It is snowing or Bart is playing hockey.
3. Consequently Anna is skiing or Bart is playing hockey.

hypotheses

1. $p \vee \neg r$
2. $r \vee q$

propositions

- p Anna is skiing
 q Bart is playing hockey
 r it is snowing

$$p \vee q$$

$$\neg p \vee r$$

Resolution rule

$$\therefore q \vee r$$

Consequently Anna is skiing or Bart is playing hockey

Solve the following:

Q.1: Show that

$$((RVQ) \wedge (PV-Q)) \Leftrightarrow ((RVQ) \wedge (PV-Q) \wedge (RVP))$$

is Valid argument

Logical equivalence

- Two sentences are **logically equivalent** iff true in same models: $\alpha \equiv \beta$ iff $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Proof by resolution

- Resolution refutation:
- Convert all sentences to CNF
- Negate the desired conclusion
- Apply resolution rule until either
 - Derive false (a contradiction)
 - Can't apply any more
- Resolution refutation is sound and complete
 - • If we derive a contradiction, then the conclusion follows from the axioms
 - • If we can't apply any more, then the conclusion cannot be proved from the axioms

CNF

- A sentence is in Conjunctive Normal Form (CNF) if it is a conjunction of clauses, each clause being a disjunction of literals
- Examples:

$$(A \vee B) \wedge (C \vee D \vee J) \wedge (E \vee G)$$

Conversion to CNF

$$P \leftrightarrow (Q \vee R)$$

1. Eliminate $\leftrightarrow$, replacing $\alpha \leftrightarrow \beta$ with $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$.

$$(P \rightarrow (Q \vee R)) \wedge ((Q \vee R) \rightarrow P)$$

2. Eliminate $\rightarrow$, replacing $\alpha \rightarrow \beta$ with $\neg \alpha \vee \beta$.

$$(\neg P \vee Q \vee R) \wedge (\neg(Q \vee R) \vee P)$$

3. Move $\neg$ inwards using de Morgan's rules and double-negation:

$$(\neg P \vee Q \vee R) \wedge ((\neg Q \wedge \neg R) \vee P)$$

4. Apply distributivity law ($\wedge$ over $\vee$) and flatten:

$$(\neg P \vee Q \vee R) \wedge (\neg Q \vee P) \wedge (\neg R \vee P)$$

Propositional Resolution Example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

So let's just do a proof. Let's say I'm given “P or Q”, “P implies R” and “Q implies R”. I would like to conclude R from these three axioms. I'll use the word "axiom" just to mean things that are given to me right at the moment.

Propositional Resolution example

- $P \vee Q$

We start by converting this first sentence into conjunctive normal form. We don't actually have to do anything. It's already in the right form.

Propositional Resolution example

- $P \vee Q$

We start by converting this first sentence into conjunctive normal form. We don't actually have to do anything. It's already in the right form.

Propositional Resolution Example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2

We can apply resolution to lines 1 and 2, and get “Q or R” by resolving away P.

Propositional Resolution Example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

And we can take lines 2 and 4, resolve away R, and get “not P.”

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4

Propositional Resolution Example

Prove R

1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

And finally, resolving away R in lines 4 and 8, we get the empty clause, which is false. We'll often draw this little black box to indicate that we've reached the desired contradiction.

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	Negated conclusion
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	•	4,8

Example Problem

Convert to CNF

Prove R

1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

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Here's an example problem. Stop and do the conversion into CNF before you go to the next slide.

Using PL prove D from A,B,C

A. $P \Rightarrow (Q \Leftrightarrow R)$

B. $\neg(Q \Leftrightarrow R)$

C. $(S \wedge Q) \Rightarrow P$

D. $\neg P \wedge (S \Rightarrow \neg Q)$

Forward and backward chaining

- **Horn Form** (restricted)
KB = **conjunction** of **Horn clauses**
 - Horn clause =
 - proposition symbol; or
 - (conjunction of symbols) $\Rightarrow$ symbol
 - E.g., $C \wedge (B \Rightarrow A) \wedge (C \wedge D \Rightarrow B)$
 -
- **Modus Ponens** (for Horn Form): complete for Horn KBs
-

$$\alpha_1, \dots, \alpha_n,$$

$$\alpha_1 \wedge \dots \wedge \alpha_n \Rightarrow \beta$$

$$\beta$$

- Can be used with **forward chaining** or **backward chaining**.
- These algorithms are very natural and run in **linear** time
-

Forward chaining

- Idea: fire any rule whose premises are satisfied in the *KB*,
 - add its conclusion to the *KB*, until query is found

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

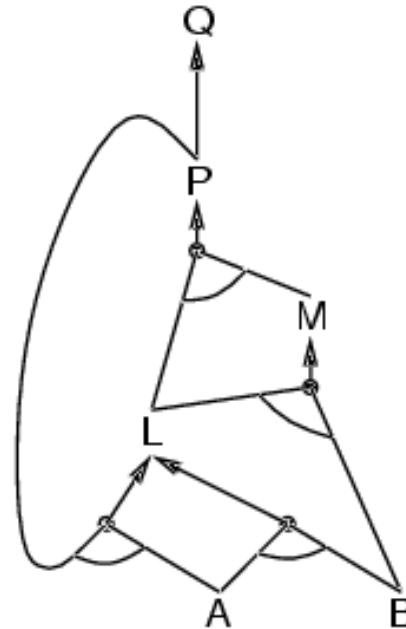
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining algorithm

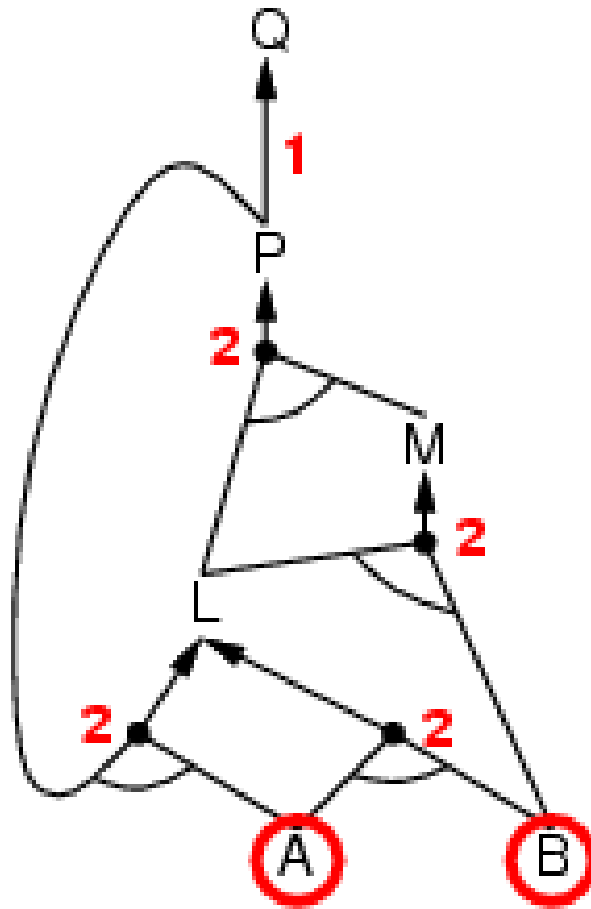
```
function PL-FC-ENTAILS?(KB, q) returns true or false
  local variables: count, a table, indexed by clause, initially the number of premises
                  inferred, a table, indexed by symbol, each entry initially false
                  agenda, a list of symbols, initially the symbols known to be true

  while agenda is not empty do
    p ← POP(agenda)
    unless inferred[p] do
      inferred[p] ← true
      for each Horn clause c in whose premise p appears do
        decrement count[c]
        if count[c] = 0 then do
          if HEAD[c] = q then return true
          PUSH(HEAD[c], agenda)

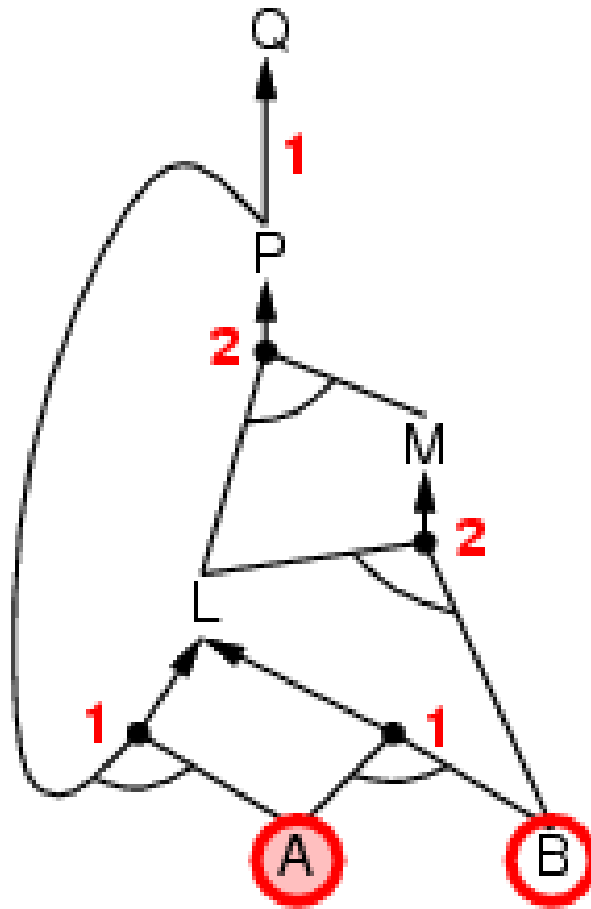
  return false
```

- Forward chaining is sound and complete for Horn KB
-

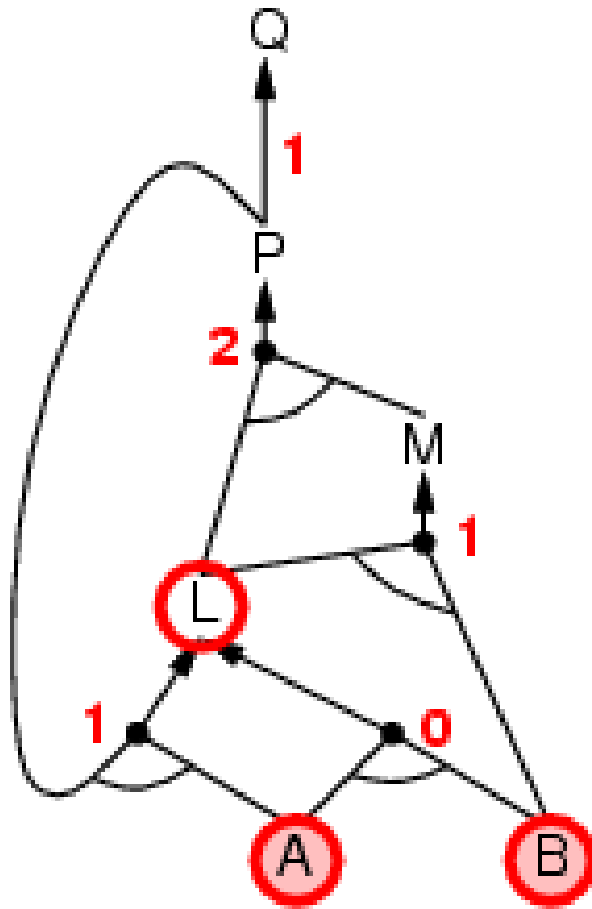
Forward chaining example



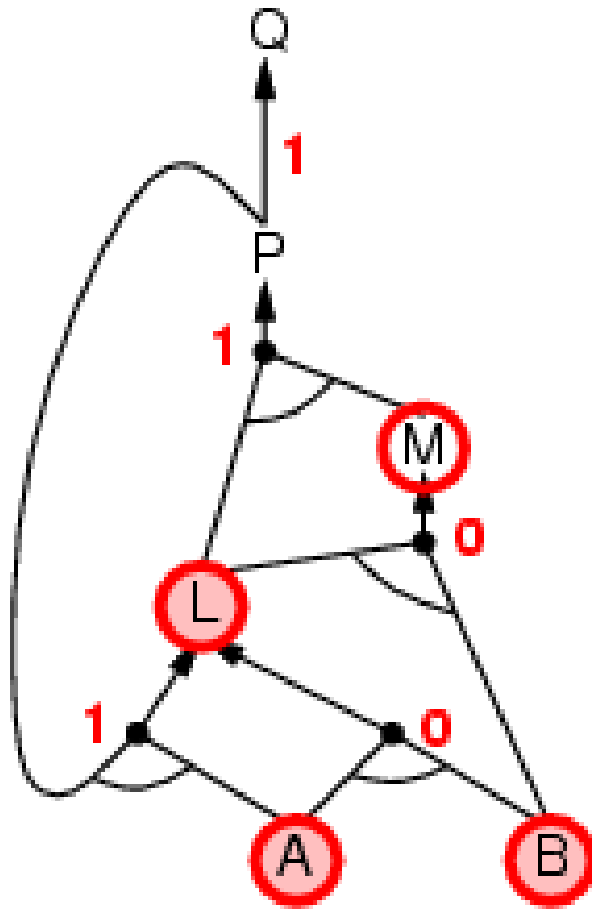
Forward chaining example



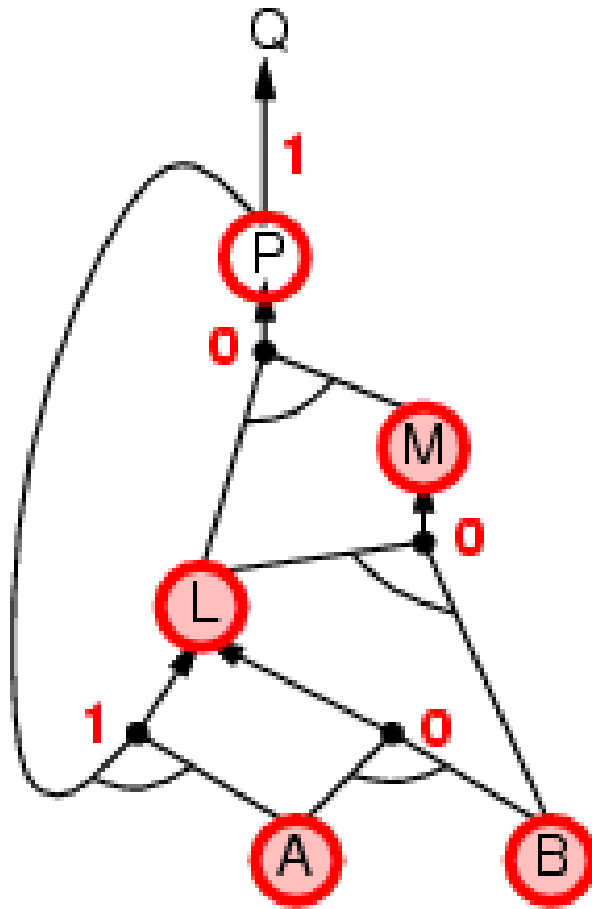
Forward chaining example



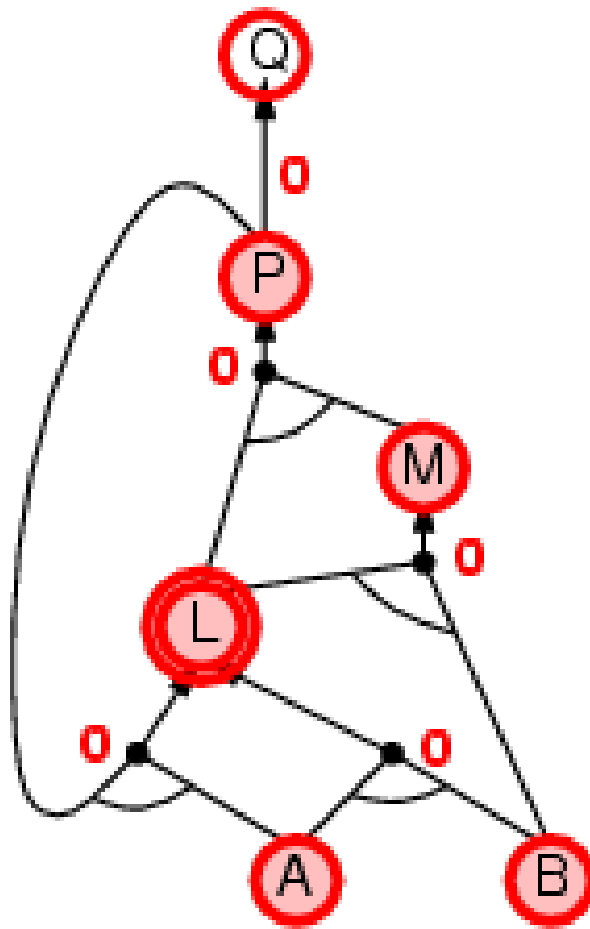
Forward chaining example



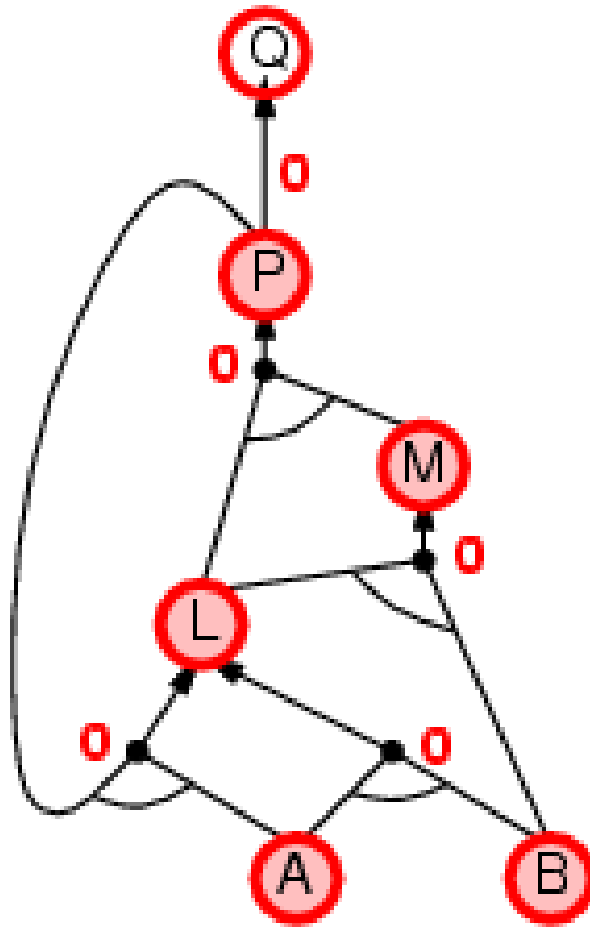
Forward chaining example



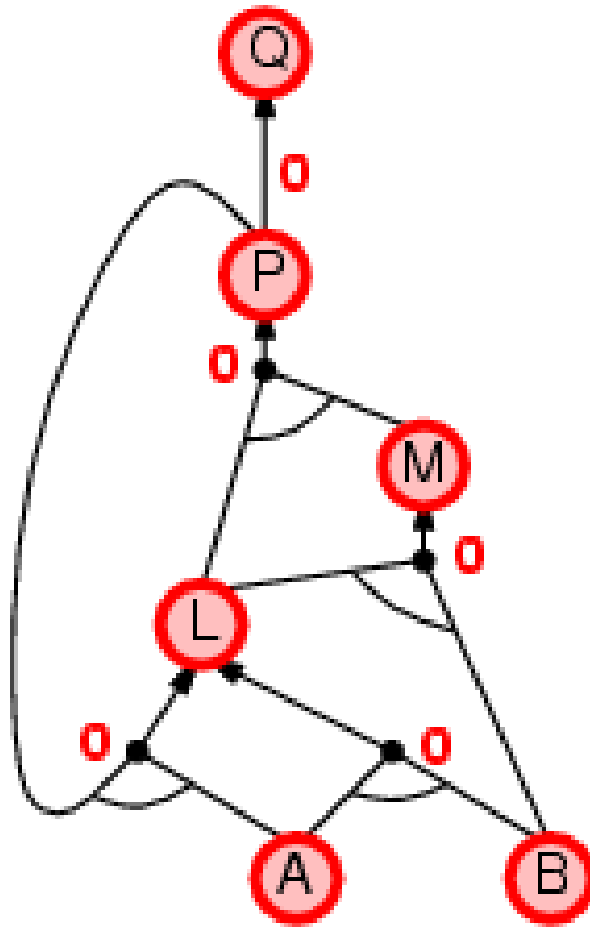
Forward chaining example



Forward chaining example



Forward chaining example



Proof of completeness

- FC derives every atomic sentence that is entailed by KB
- 1. FC reaches a **fixed point** where no new atomic sentences are derived
 - 2.
 2. Consider the final state as a model m , assigning true/false to symbols
 - 3.
 3. Every clause in the original KB is true in m
 - 4.
$$a_1 \wedge \dots \wedge a_k \Rightarrow b$$
 4. Hence m is a model of KB
 - 5.
 5. If $KB \models q$, q is true in **every** model of KB , including m
 - 6.

Backward chaining

Idea: work backwards from the query q :

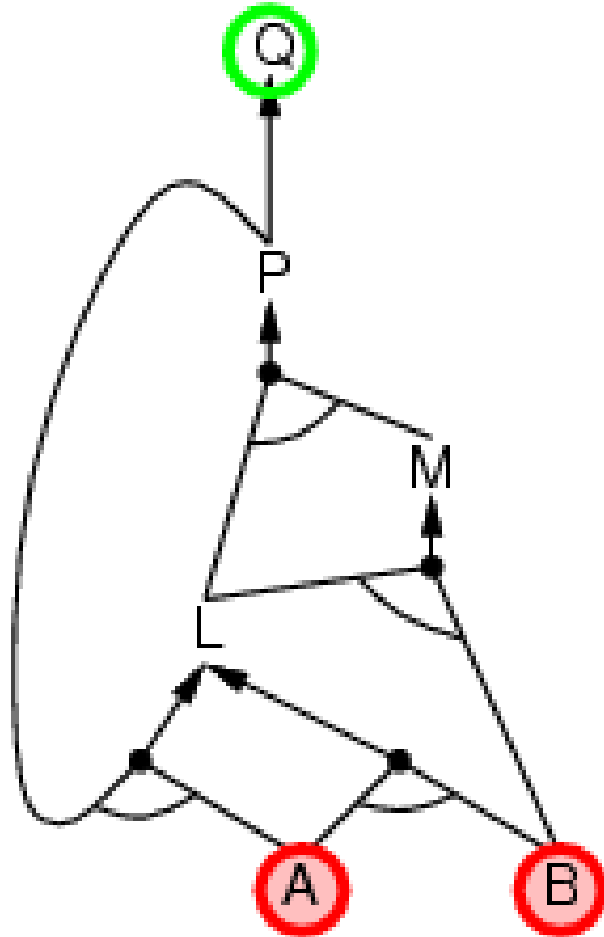
to prove q by BC,
 check if q is known already, or
 prove by BC all premises of some rule concluding q

Avoid loops: check if new subgoal is already on the goal stack

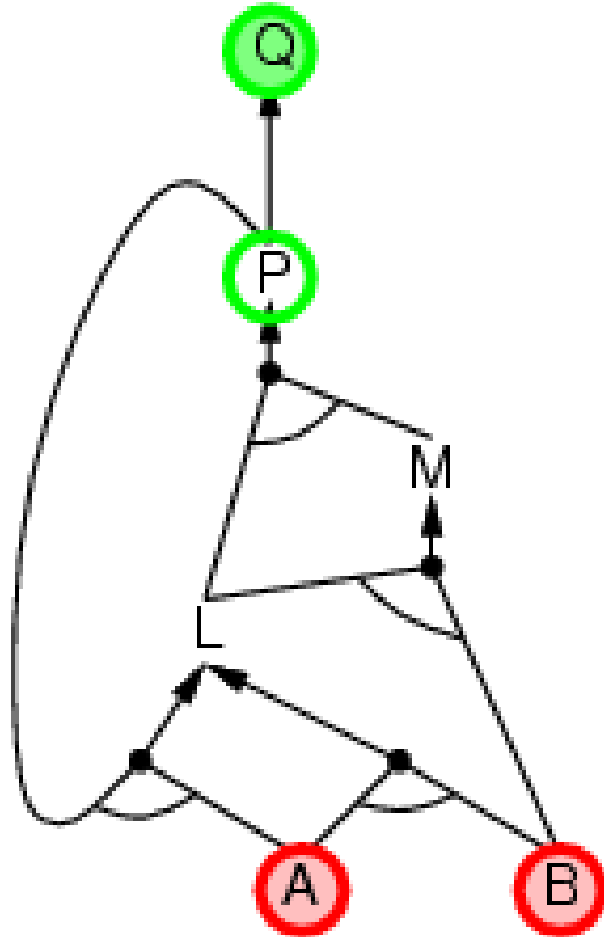
Avoid repeated work: check if new subgoal

1. has already been proved true, or
- 2.
2. has already failed

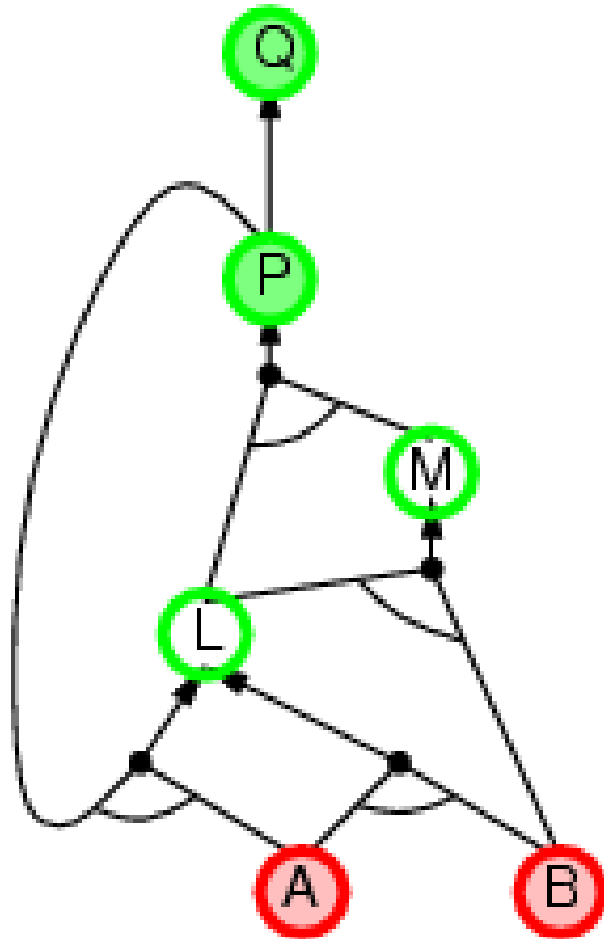
Backward chaining example



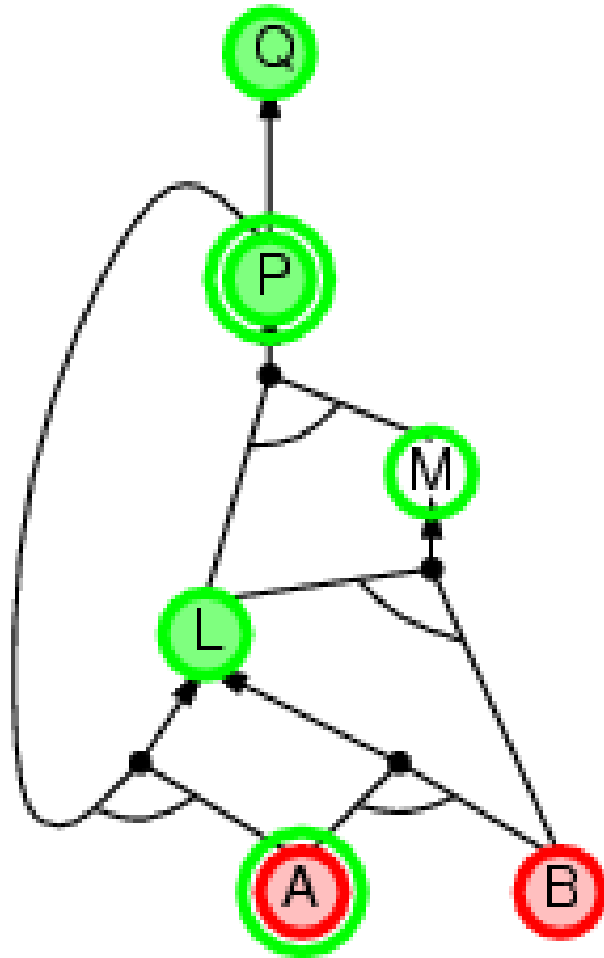
Backward chaining example



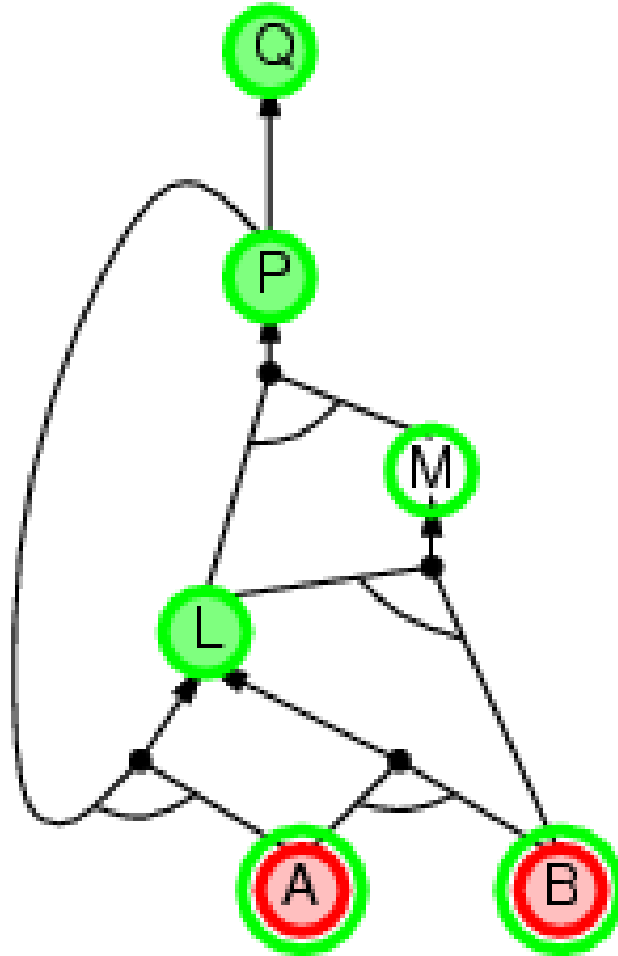
Backward chaining example



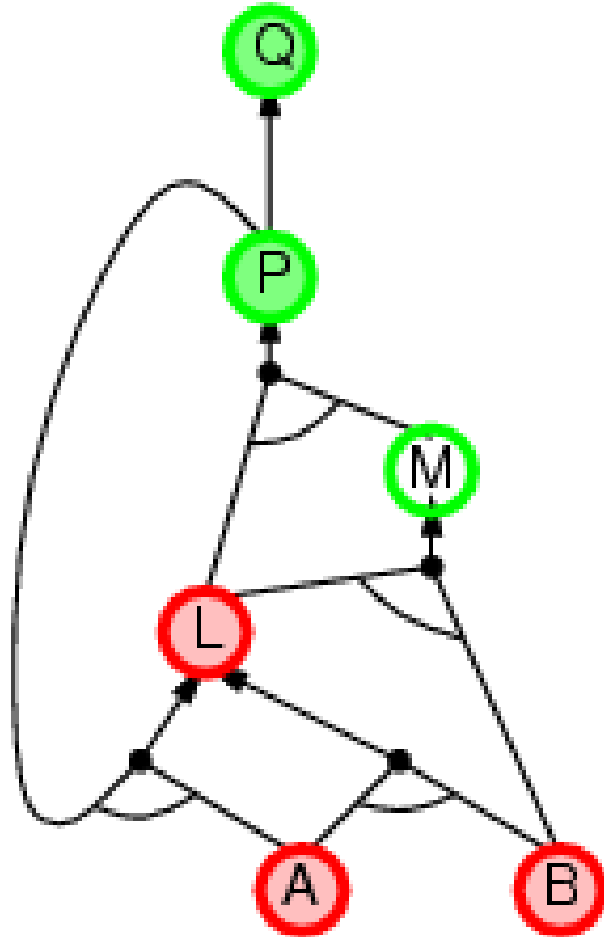
Backward chaining example



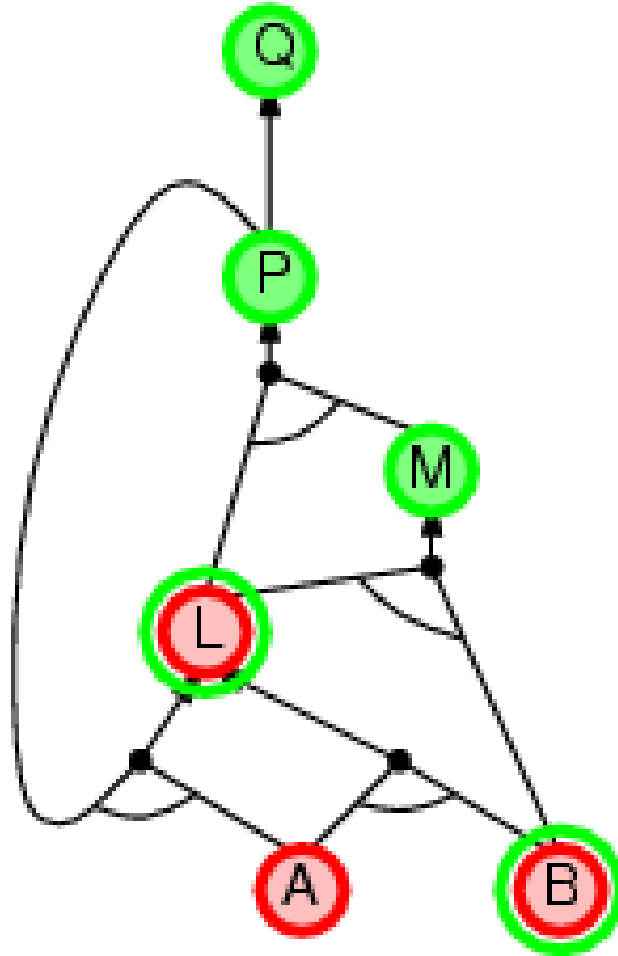
Backward chaining example



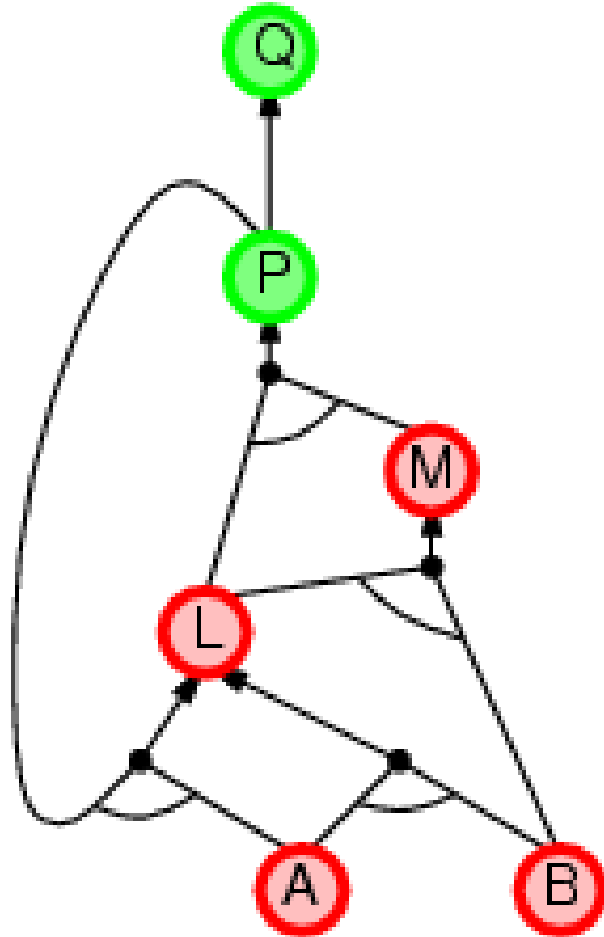
Backward chaining example



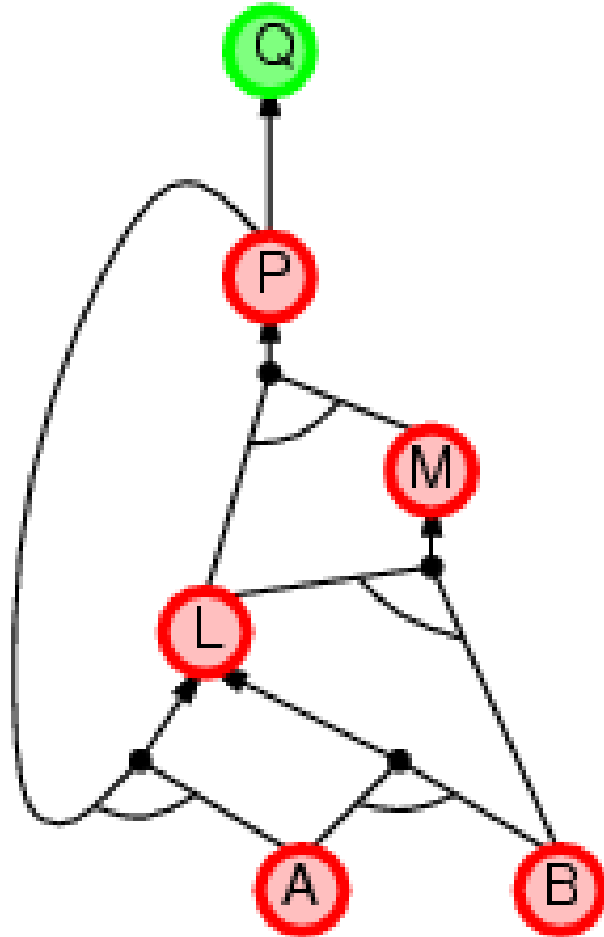
Backward chaining example



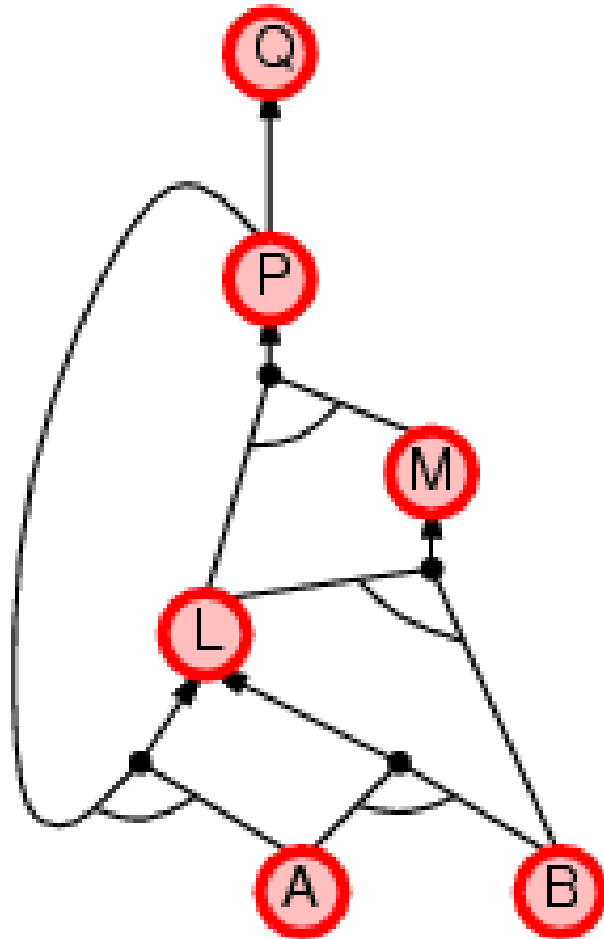
Backward chaining example



Backward chaining example



Backward chaining example



Forward vs. backward chaining

- FC is **data-driven**, automatic, unconscious processing,
 - e.g., object recognition, routine decisions
 -
- May do lots of work that is irrelevant to the goal
- BC is **goal-driven**, appropriate for problem-solving,
 - e.g., Where are my keys? How do I get into a PhD program?
- Complexity of BC can be **much less** than linear in size of KB
-

Convert following sentences into propositional logic and prove that the statement Alice goes to college is true

1. Alice likes maths and she likes stories
2. If she likes maths, she likes algebra
3. If she likes algebra and likes physics she will go to college
4. She does not like stories or she likes physics
5. She does not like chemistry and history